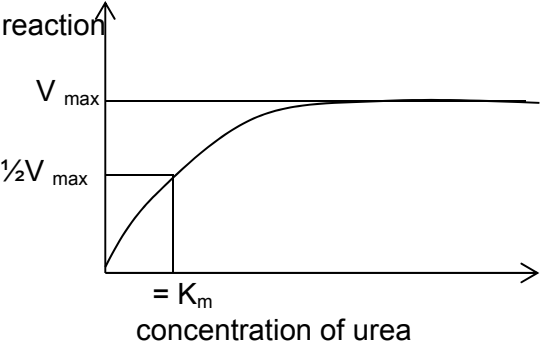


1(a)(i)	weigh / measure (out) / take / put / add / use/ AW, 6 (.01) g urea ; add 500 cm ³ (distilled / deionised) water (and stir until dissolved) ;	2
1(a)(ii)	<i>idea of</i> removing (a known and) same volume of urea solution (with second dilution taken from first dilution, etc.) at each stage of dilution ; <i>idea of</i> adding (a known and) same volume of (distilled) water at each stage of dilution ; A as a diagram showing sequence of dilution A as a table showing volumes (and concentrations) A proportional dilution for max 1 (ecf) if 4 dilutions correctly gained	2
1(b)(i)	<i>independent</i> temperature ; <i>dependent</i> conductivity (of enzyme and substrate / ions / solution) ; A in conductivity units	2
1(b)(ii)	substituting, the active enzyme / urease, by an unreactive substance (at all temperatures) A e.g. boiled or denatured enzyme / water in place of enzyme / 0.0M enzyme / (solution of) urea without urease or substituting, urea / substrate, with water ; I 'use (distilled) water' unqualified A e.g. water in place of substrate / 0.0M substrate / (solution of) urease without urea	1

Question	Answer	Marks
1(b)(iii)	<i>max 5 if mp7 not given</i> <i>any 6 from:</i> <i>ref. to a suitable range of at least 5 temperatures ; A any 5 in the range 10–70 °C</i> <i>ref. to using suitable apparatus (to incubate enzyme and urea solutions at constant temperature(s)) ; e.g. (thermostatically controlled) water baths / incubators / thermostatically or temperature controlled room / (magnetic stirrer) hotplate A beaker plus hot water as water-bath I air conditioning</i> <i>ref. to using same volume of urease each time ; total volume must not exceed 30 cm³</i> <i>ref. to using same volume of (each) urea concentration ; total volume must not exceed 30 cm³</i> <i>ref. to using (same volume of) buffer to maintain a constant pH ; A use buffer to control pH</i> <i>procedure:</i> <i>ref. to incubating urease and urea concentrations separately ;</i> <i>ref. to mixing urea and urease solutions (on the magnetic stirrer) and immersing (conductivity) probe ; I 'probe is used' unqualified</i> <i>ref. to taking reading (from meter) at same time (for each solution / temperature) ; A any stated time from 0 s to 5 min A take reading immediately / AW A 'use meter to measure rate of reaction' if time context correct</i> <i>ref. to testing each of the concentrations (of urea) at each temperature ;</i> <i>ref. to a min. of 3 replicates / repeats and a mean / find anomalies ;</i>	6

Question	Answer	Marks
1(b)(iii)	11 <i>ref. to suitable hazard and precaution ;</i> A low risk experiment I medium risk experiment R no risk / high risk A urease / enzyme is allergen or irritant and wear gloves / goggles A Ammonia (given off at high temperature) is irritant / mask or gloves	
1(c)(i)	1 one curve drawn correctly ; <i>does not need to go through origin</i> A straight line up then level without curving 2 $V_{\max}$ shown ; rate of reaction 3 $\frac{1}{2}V_{\max}$ shown ;  = K_m concentration of urea 4 K_m shown ; A Michaelis-Menten (constant)	4
1(c)(ii)	1 D ; 2 (temperature showing) lowest K_m or shows, greater / greatest / higher / highest, affinity (of the, enzyme / urease) for its substrate / AW ;	2